# Understanding GIS: Assessment 2 Report — Weighted Redistribution

***[Subject: GEOG71551]***
***[Submitted By:14340308]***

# Introduction

The use of geocoded social media data can often lead to incorrect interpretations when the events are assigned to large administrative areas rather than to precise locations, thus artificially clustering in centroids or boundaries, which are sometimes referred to as fake hotspots due to encoding quirks rather than being valid spatial patterns (Huck et al., 2015). Such a situation can really misguide the authorities in determining the location of the actual public interest.

The Royal Wedding tweet case from Greater Manchester demonstrates this issue because district-level geocoding creates false clusters while making spatial uncertainty hidden. The Weighted Redistribution (WR) technique provides a solution through a GIS method—probabilistically relocating points within source zones with a population density weight (Gregory, 2002; Huck et al., 2015). In this research, we use WR to depict a more believable distribution of wedding enthusiasm as a seamless intensity surface.

# Method and justification

This follows Huck et al.'s single-level Weighted Redistribution workflow, just district scale, skipping the outer "levels" loop from the paper. The goal isn't to recover exact tweet locations (impossible) but to relocate each to a plausible spot within its district, so the overall surface better reflects actual patterns and cuts out those fake hotspots from the geocoding.

That's why this report focuses on justifying each step rather than detailing the code, as required by the brief. First key choice: projecting everything to British National Grid (EPSG:27700) so areas, radii, and distances work in metres, which is essential since the splat radius r derives from polygon area and fuzziness parameter s. This keeps calculations metric and meaningful for Greater Manchester scale, avoiding any mix-up with angular units.

```
target_crs = CRS.from_epsg(27700)
if tweets.crs != target_crs:
    tweets = tweets.to_crs(target_crs)
if districts.crs != target_crs:
    districts = districts.to_crs(target_crs)
```

The redistribution stays strictly within each district polygon, so a tweet originally geocoded to Salford can't jump to Trafford, this keeps things realistic and true to the algorithm's premise of disaggregating only within the known reporting unit.

Candidate points are generated with a simple bounding-box rejection method (quick to code and easy to follow): random points get sampled inside the district's bounding box until one falls within the polygon . This student-friendly approach ensures accuracy without complex spatial sampling techniques .

The scoring process evaluates each candidate by comparing their performance against the population raster which serves as the weighting surface. The *w* parameter dials in how much the weights pull: it samples w candidates and picks the one with the top population value, so bigger *w* ramps up the bias toward dense cells and yields a more population-driven surface.

```
def relocate_weighted(point, polygon, raster, w):
    best, best_val = None, -1
    for _ in range(w):
        cand = random_point_in_polygon(polygon)
        val = raster_value(cand, raster)
        if val > best_val:
            best_val, best = val, cand
    return best
```

After shifting the "seed" point, each tweet turns into a fuzzy splat, a spread-out local distribution instead of a single sharp pixel, to capture spatial uncertainty rather than faking precision. The splat radius *r* comes from the polygon area and the ambiguity parameter *s* (0–1), so bigger districts or higher fuzziness mean broader splats.
The fuzzy kernel gets built directly for clear logic: values drop off with distance from the seed, stacking up into a smooth overall surface.

```
def splat_radius(polygon, s):
    return math.sqrt((polygon.area * s) / math.pi)

def add_splat(surface, patch, point, raster):
    row, col = rasterio.transform.rowcol(raster.transform, point.x, point.y)
    r = patch.shape[0] // 2
    for i in range(patch.shape[0]):
        for j in range(patch.shape[1]):
            rr, cc = row - r + i, col - r + j
            if 0 <= rr < surface.shape[0] and 0 <= cc < surface.shape[1]:
                surface[rr, cc] += patch[i, j]
```

Parameter choices came from a quick sweep, just as required:
tested $s \in \{0.03, 0.05\}$ and $w \in \{25, 50, 100\}$ via visual checks on saved outputs. For me, $w$=50 gave solid population-guided redistribution without getting too peaky (like $w$=100 did), and $s$=0.05 yielded a clear continuous surface without over-smoothing district edges (unlike lower $s$). These felt like a balanced, justifiable pick for the client map, backed by the exploration figures rather than random defaults.

# Results

In Figure 1, a significant flaw with geocoding at the district level is displayed, with tweets being clustered at the same places, resulting in the creation of misleading but very colorful hotspots. Similar aggregation artefacts crop up often in spatial analysis, especially when point data gets boxed into coarse areal units (Goldberg and Wilson, 2011) .

Figure 4, after weighted redistribution, exposes an entirely new spatial pattern. Point clusters that were sharp disappear and attract a continuous intensity that is limited by the district's edges and follows the areas of higher population densities because relocation stays within district boundaries, intensity can break abruptly near borders. This happens because of the algorithm at work: tweets shift probabilistically within their original districts, guided by

population weights, so Greater Manchester's urban cores stand out, sparse areas fade, perfectly matching dasymetric principles that use extra data to refine coarse aggregates (Eicher and Brewer, 2001).

Parameter sweeps validate the selections really well: the low w was too random in its scattering, the high w was unrealistically spiking; the small s brought back point artifacts while the large s was blurring the key details. Visual confirmations on Figures 2 and 3 pinpointed w=50 and s=0.05 as being both even and defensible, thus following the best practice of sensitivity analysis (Leyk et al., 2019).

***Figure 1. Raw tweet locations geocoded at district level***

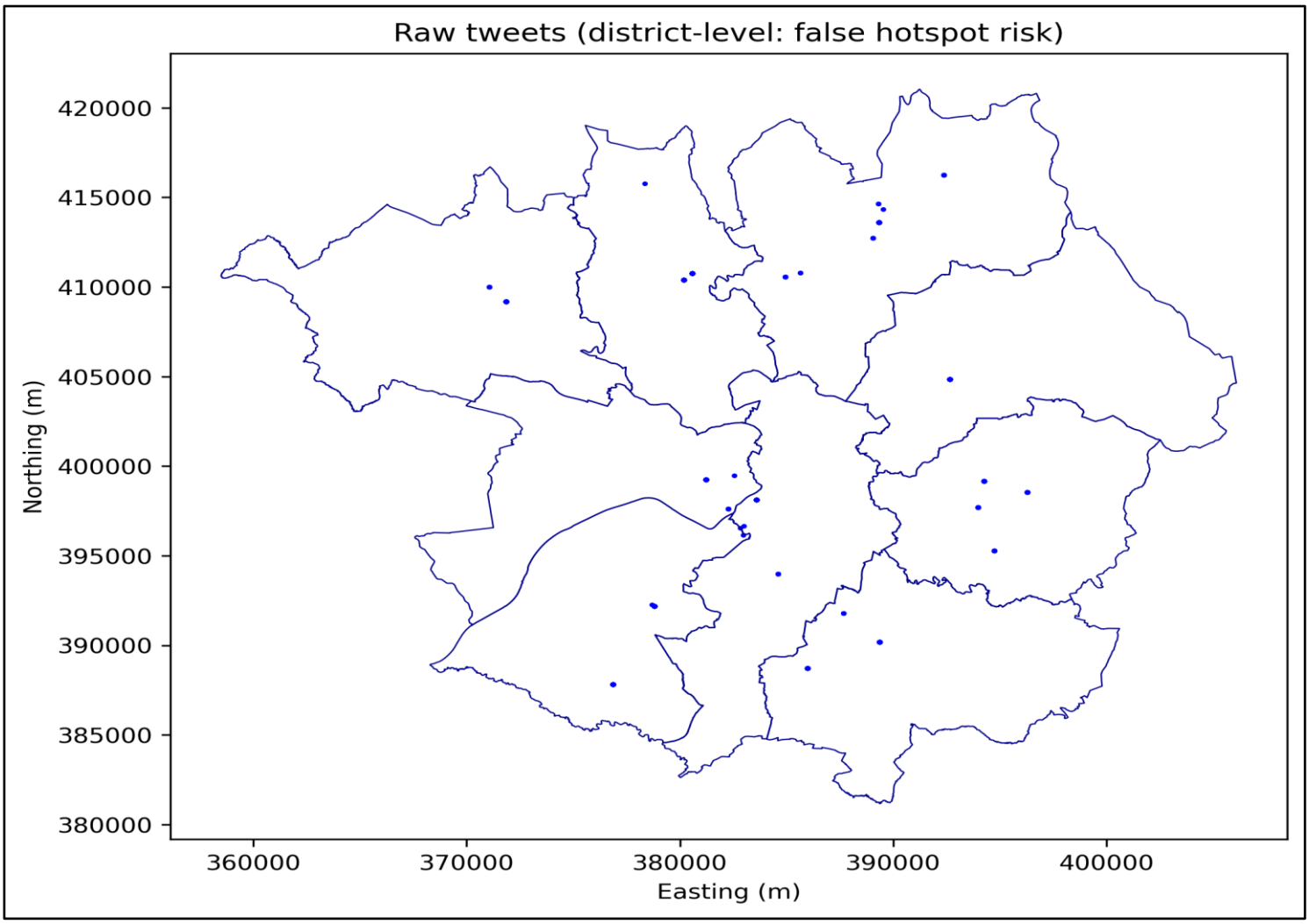
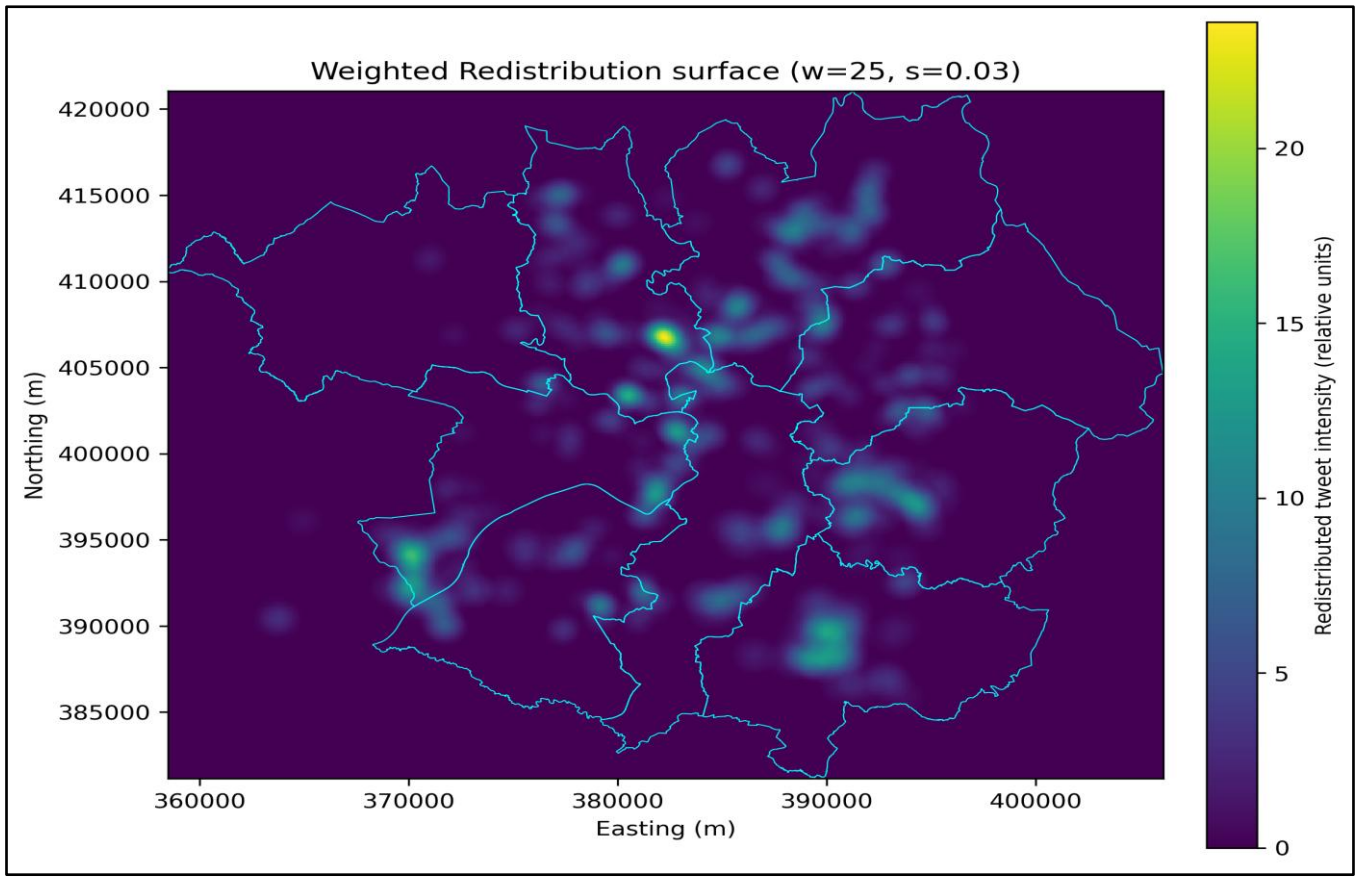


***Figure 2. Effect of varying the fuzziness parameter (s) on the weighted redistribution surface (changes in spatial spread)***



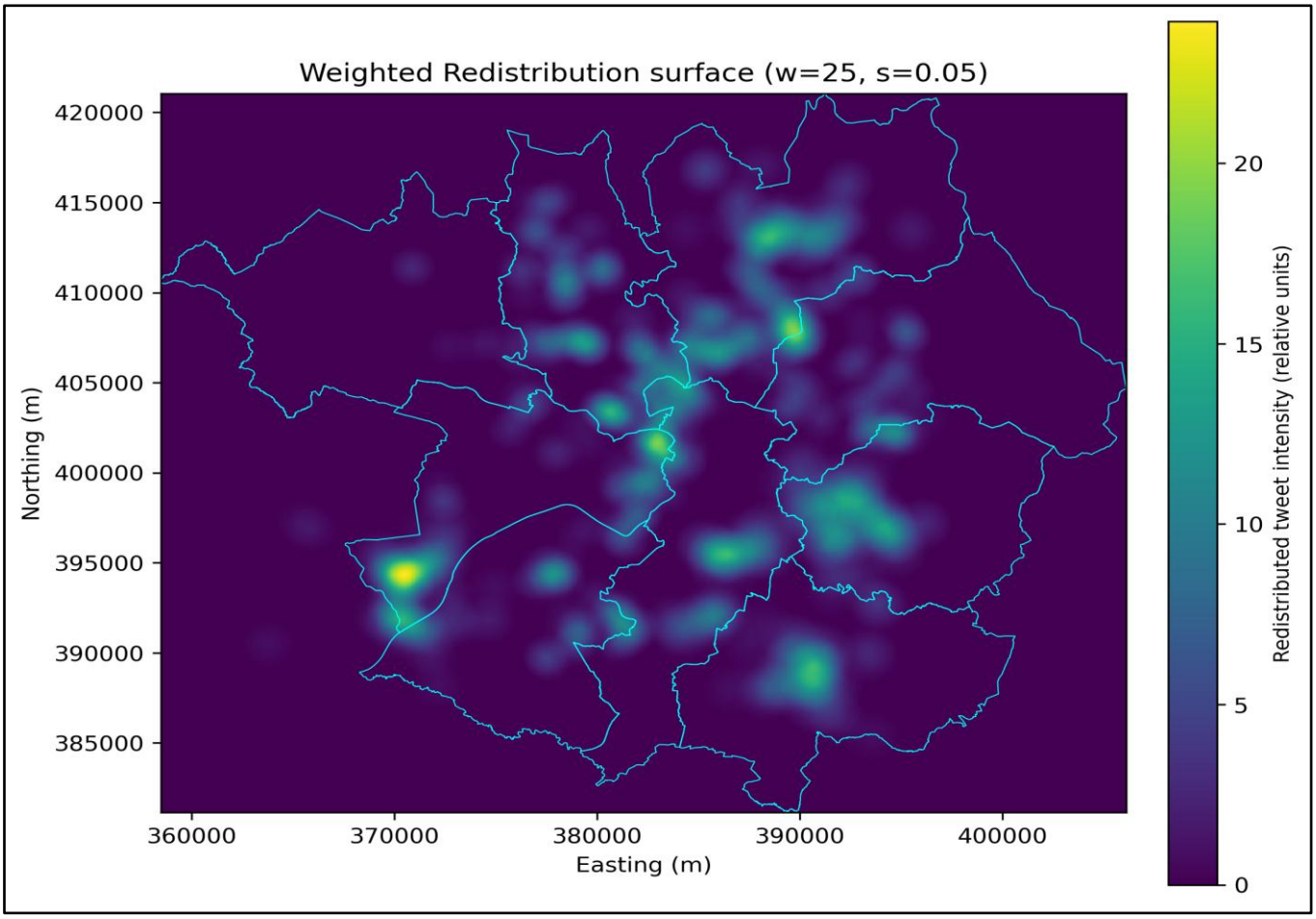



***Figure 3. Effect of varying the weighting parameter (w) on redistribution intensity (increasing dominance of the population surface).***

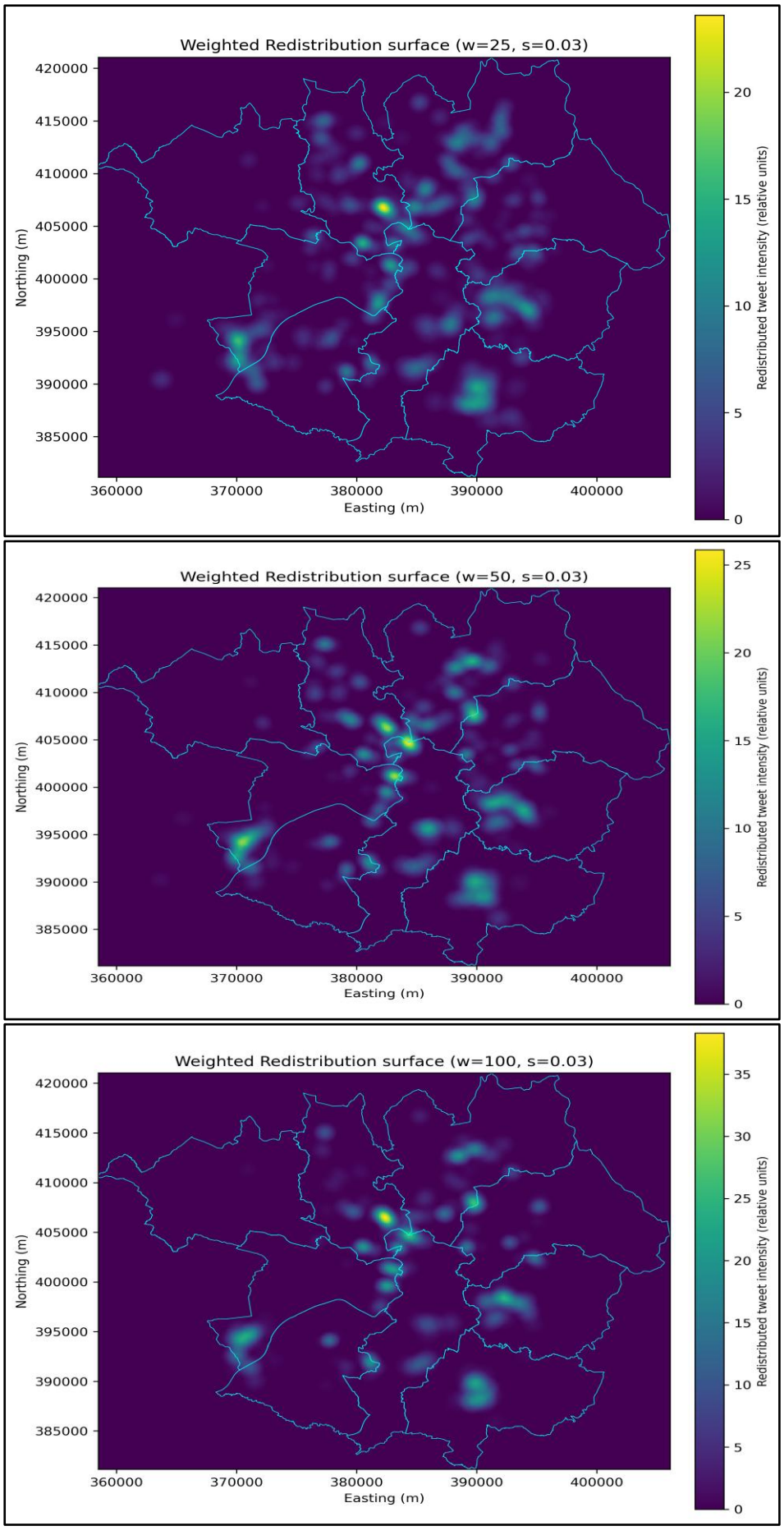
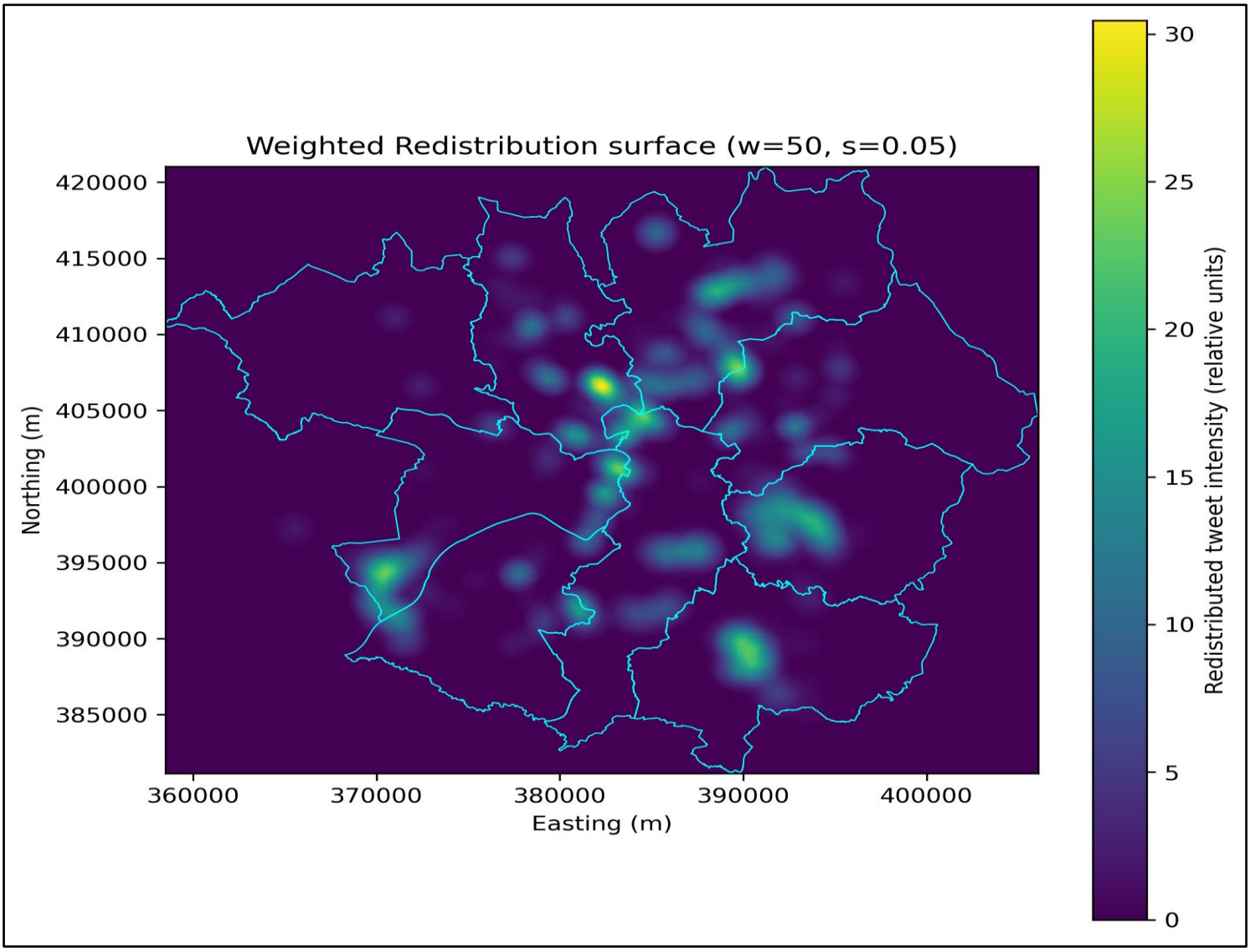


***Figure 4. Final weighted redistribution surface (w = 50, s = 0.05)***




# Limitations and Improvements

There are a number of limitations which should be pointed out while interpreting the surface of the weighted redistribution. The very nature of its meaning assumes that population density is the only proxy for all the tweeting activities within the districts. Besides that, it does not consider the movement of people to and from the event spots, which is a major downside of one-layer dasymetric techniques (Eicher and Brewer, 2001).

Stochastic relocation also means that results can not be completely reproduced unless a specific seed is used; there will be minor local shifts between the different runs. The parameters w and s were not optimized by rigorous means but rather through visual checks that fit the exploratory focus, while the boundary limits impede the flow of tweets and, thus, may disrupt continuity in those areas (Huck et al., 2015). Time constraints did not allow for the use of better weighting options, proper sensitivity tests, and validation against accurate geocoded data (Leyk et al., 2019).

# Conclusion

This study applied weighted redistribution to handle spatial uncertainty in district-level Royal Wedding Twitter data across Greater Manchester. Raw points revealed false hotspots from admin encoding that distort public interest patterns, limiting spatial use .

Probabilistic redistribution within districts, weighted by population density, produced a plausible continuous intensity surface respecting geographic bounds. Parameter tests showed *w* and *s* matter: low *w* scattered randomly, high *w* peaked sharply; small *s* added point artefacts, large *s* over-smoothed. *w*=50, *s*=0.05 balanced realism and clarity via visual sensitivity checks.
View the surface as exploratory relative interest, not exact locations. It upgrades raw maps for clients if limits are stated upfront

# References

Huck, J., Whyatt, D. and Coulton, P., 2015. Visualizing patterns in spatially ambiguous point data. *Journal of Spatial Information Science*, *2015*(10), pp.47-66. Available at: http://dx.doi.org/10.5311/JOSIS.2015.10.211 (Accessed: 17 December 2025)

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Goldberg, D.W., Wilson, J.P. and Knoblock, C.A., 2007. From text to geographic coordinates: the current state of geocoding. *URISA journal*, *19*(1), pp.33-46. Available at: https://www.researchgate.net/profile/John-Wilson-14/publication/281748610_From_text_to_geographic_coordinates_The_current_state_of_geocoding/links/57478f2708aef66a78b07e2f/From-text-to-geographic-coordinates-The-current-state-of-geocoding.pdf (Accessed: 25 December 2025)

Leyk, S., Gaughan, A.E., Adamo, S.B., De Sherbinin, A., Balk, D., Freire, S., Rose, A., Stevens, F.R., Blankespoor, B., Frye, C. and Comenetz, J., 2019. The spatial allocation of population: a review of large-scale gridded population data products and their fitness for use. *Earth System Science Data*, *11*(3), pp.1385-1409. Available at: https://doi.org/10.5194/essd-11-1385-2019 (Accessed: 26 December 2025)

# Statement of AI usage

ChatGPT (OpenAI) was used mainly for limited language support consisting of text clarity and conciseness improvement, and also for high-level conceptual discussion about some GIS principles. Not in a way to generate code, code comments, or make any methodological decisions.

NotebookLM (Google) was utilized not only for sorting and reviewing the course materials, research papers, and assessment documents provided but also for help in extracting the main ideas from the uploaded sources. I did all the interpretation, choice of methods, and writing independently. It was mainly used for conceptual clarification.

Grammarly was used to check the final report text for proofreading, spelling, and grammatical correction.